Information-Theoretic Analysis of Instruction Grounding in VLA Models

1. Preliminaries and Notation

Consider a Vision-Language-Action (VLA) model that learns a policy $\pi_\theta(a_t \mid v_t, l, p_t)$, where at each timestep t:

- $V_t \in \mathcal{V}$: visual observation (image),
- $L \in \mathcal{L}$: natural language instruction,
- $P_t \in \mathcal{P}$: proprioceptive state,
- $A_t \in \mathcal{A}$: continuous action.

For notational compactness, we absorb P_t into V_t and write the observation as $O_t = (V_t, L)$. The model is trained via supervised fine-tuning (SFT) on expert demonstrations $\mathcal{D} = \{(v_t, l, a_t^*)\}_{t=1}^N$ by minimising:

$$\mathcal{L}(\theta) = \mathbb{E}_{(V,L,A)\sim\mathcal{D}} \left[-\log \pi_\theta(A \mid V, L)\right] = \mathbb{E}\left[H_\theta(A \mid V, L)\right].$$

We omit the timestep subscript where it is clear from context.

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2. The Instruction Redundancy Bound

2.1 Core Inequality

By the definition of conditional mutual information:

$$I(A; L \mid V) = H(A \mid V) - H(A \mid V, L).$$

Simultaneously, a standard property of mutual information gives:

$$I(A; L \mid V) \leq H(L \mid V), \tag{1}$$

since $I(A; L \mid V) = H(L \mid V) - H(L \mid V, A)$ and $H(L \mid V, A) \geq 0$. Rearranging Eq. (1):

$$\boxed{H(A \mid V) - H(A \mid V, L) \leq H(L \mid V).} \tag{2}$$

Interpretation. The left-hand side of Eq. (2) is the *information gain* about the action that the language instruction provides beyond what is already available from vision. This gain is upper-bounded by the *conditional entropy of the instruction given the visual observation*. When $H(L \mid V)$ is small—i.e., the instruction is largely predictable from the image—the language channel can contribute at most a negligible amount of information about the correct action.

2.2 When Does $H(L \mid V)$ Collapse?

In standard VLA training, each expert trajectory $\tau = (v_1, a_1, v_2, a_2, \ldots, v_N, a_N)$ is paired with a single, fixed task-level instruction l (e.g., *"pick up the red cup and place it on the plate"*). Two consequences follow:

1. **Temporal constancy.** Across all N timesteps of a trajectory, the instruction l is identical. The visual observations $\{v_t\}_{t=1}^{N}$, however, vary with the evolving scene. This means the per-timestep conditional entropy $H(L \mid V_t)$ can be decomposed as:

$$H(L \mid V_t) = H(L) - I(L; V_t). \tag{3}$$

Since the task-level instruction l uniquely determines (and is determined by) the task identity, and the visual scene at any timestep t is strongly correlated with that identity (the robot is visibly manipulating specific objects in a specific configuration), we have $I(L; V_t) \approx H(L)$ for most t, implying $H(L \mid V_t) \approx 0$.

2. **Cross-trajectory leakage.** Even across different trajectories, if the dataset contains a moderate number of distinct tasks (e.g., 10 tasks in LIBERO-Long), the task instruction is low-entropy a priori: $H(L) = \log |\mathcal{T}|$ is small. Once the model observes the initial scene, the instruction becomes almost fully determined. Concretely, for a LIBERO suite with $|\mathcal{T}| = 10$ tasks and visually distinguishable initial configurations:

$$H(L \mid V_0) \ll H(L) = \log_2 10 \approx 3.32 \text{ bits}.$$

Combining both effects, across the training distribution $\mathcal{D}$:

$$H(L \mid V) \approx 0 \quad \Longrightarrow \quad I(A; L \mid V) \approx 0. \tag{4}$$

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3. Gradient Analysis: Language Parameters Become Dormant

We now show that the information-theoretic bound in Eq. (2) has a direct consequence on the gradient dynamics of the model parameters responsible for processing language.

3.1 Model Decomposition

Suppose the VLA model factorises its parameters as $\theta = (\theta_V, \theta_L, \theta_{VL})$, where θ_V processes visual tokens, θ_L processes language tokens, and θ_{VL} fuses the two modalities to produce the action distribution. The SFT loss can be written as:

$$\mathcal{L}(\theta) = \mathbb{E}_{(V,L,A)} \left[-\log \pi_\theta(A \mid V, L) \right].$$

3.2 Optimal Model Under Low $H(L \mid V)$

Consider two models:

- $\pi^*(A \mid V, L)$: the Bayes-optimal policy conditioned on both V and L,
- $\pi^*_V(A \mid V)$: the Bayes-optimal policy conditioned on V alone.

The expected log-likelihood gap between them is:

$$\mathbb{E}\left[\log \pi^*(A \mid V, L) - \log \pi^*_V(A \mid V)\right] = D_{\mathrm{KL}}[p(A \mid V, L) \,\|\, p(A \mid V)]_{\mathbb{E}_{V,L}}$$

$$= I(A; L \mid V) \leq H(L \mid V). \tag{5}$$

When $H(L \mid V) \approx 0$, the two models achieve essentially the same expected log-likelihood. This means the loss landscape has a basin of near-optimal solutions where $\pi_\theta(A \mid V, L) \approx \pi_\theta(A \mid V)$—that is, solutions that effectively ignore L.

3.3 Gradient Vanishing for θ_L

More formally, consider the gradient of the loss with respect to θ_L:

$$\nabla_{\theta_L} \mathcal{L} = -\mathbb{E}_{(V,L,A)} \left[\nabla_{\theta_L} \log \pi_\theta(A \mid V, L) \right]. \tag{6}$$

We can decompose the score function via the chain rule through the model's internal representation. Let $h_L = f_{\theta_L}(L)$ denote the language representation. Then:

$$\nabla_{\theta_L} \log \pi_\theta(A \mid V, L) = \frac{\partial \log \pi_\theta}{\partial h_L} \cdot \frac{\partial h_L}{\partial \theta_L}. \tag{7}$$

The term $\frac{\partial \log \pi_\theta}{\partial h_L}$ measures how much the action log-likelihood changes when the language representation shifts. When the model has converged to a solution where $\pi_\theta(A \mid V, L) \approx \pi_\theta(A \mid V)$—which is a near-optimal solution under the condition $H(L \mid V) \approx 0$ as shown in §3.2—the action distribution is approximately invariant to h_L. Therefore:

$$\left\| \frac{\partial \log \pi_\theta}{\partial h_L} \right\| \approx 0 \quad \Longrightarrow \quad \|\nabla_{\theta_L} \mathcal{L}\| \approx 0. \tag{8}$$

Result. Under low $H(L \mid V)$, gradient-based optimisation drives the model toward a solution where the language pathway contributes negligibly to the action prediction. The VLA model *degenerates* into a vision-action (VA) model: $\pi_\theta(A \mid V, L) \to \pi_\theta(A \mid V)$.

3.4 Empirical Signature

This degeneration manifests as the following testable prediction: ablating the language instruction at inference time (e.g., replacing L with a null or random string) should yield task success rates comparable to those achieved with the correct instruction. Our experiments on LIBERO confirm this prediction—models trained with fixed task-level instructions maintain high success rates even when the instruction is removed entirely.

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4. Restoring Instruction Sensitivity via Substep Decomposition

4.1 The APD Intervention

The Action Plan Dataset (APD) decomposes a fixed task instruction l into a sequence of time-varying substep instructions $\{l_t^{\text{sub}}\}_{t=1}^N$, where l_t^{sub} describes the *current subgoal* at timestep t (e.g., *"grasp the handle of the drawer"* rather than *"open the top drawer and put the bowl inside"*).

This transforms the training distribution from (V_t, l, A_t) to $(V_t, l_t^{\text{sub}}, A_t)$, with a crucial consequence for the conditional entropy.

4.2 $H(L^{\text{sub}} \mid V)$ Is Substantially Higher

Under the substep regime, the instruction is no longer constant across a trajectory. At each timestep t, the substep instruction l_t^{sub} depends on the *progress* through the task plan, which is only partially observable from a single image v_t. Specifically:

1. **Temporal variation.** Within a single trajectory, the substep instruction changes K times (where K is the number of subgoals), introducing genuine per-timestep variation in L_t^{sub} even when V_t evolves smoothly. This breaks the degeneracy $I(L; V_t) \approx H(L)$.
2. **Visual ambiguity.** Multiple subgoals may correspond to visually similar scenes. For example, the robot arm may be near the same object for *"grasp the cup"* and *"release the cup"*—the correct action depends critically on which substep instruction is active. Formally:

$$H(L_t^{\text{sub}} \mid V_t) \gg H(L \mid V_t) \approx 0. \tag{9}$$

3. **Increased action space per visual state.** Since the same visual observation can now be paired with different substep instructions that require different actions, the conditional mutual information is no longer negligible:

$$I(A_t; L_t^{\text{sub}} \mid V_t) \leq H(L_t^{\text{sub}} \mid V_t), \tag{10}$$

where the bound is now *non-trivially loose*—i.e., there is meaningful information to be extracted.

4.3 Gradient Reactivation

Revisiting the gradient analysis of §3.3 under the substep regime: since $\pi_\theta(A \mid V, L^{\text{sub}}) \neq \pi_\theta(A \mid V)$ at the optimal solution (the same image now requires different actions for different substep instructions), the partial derivative $\frac{\partial \log \pi_\theta}{\partial h_L}$ is no longer vanishing. The language pathway receives non-trivial gradients, and the model is forced to utilise the instruction to minimise the loss:

$$H(L^{\text{sub}} \mid V) \gg 0 \implies \|\nabla_{\theta_L} \mathcal{L}\| \gg 0. \tag{11}$$

The model can no longer achieve a low loss by relying on vision alone; it must *attend to and condition on* the substep instruction.

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5. Reinforcing Instruction Grounding via Contrastive RL

5.1 Motivation

While APD-based SFT restores the necessary condition for instruction grounding (non-trivial $H(L^{\text{sub}} \mid V)$), the SFT objective alone does not explicitly *penalise* instruction-ignoring behaviour. The maximum likelihood objective rewards generating the correct action given the correct instruction, but it does not verify that a *different* instruction would have produced a *different* action.

5.2 The Contrastive Reward

We introduce a contrastive reward R_{contrast} within an RL post-training stage (GRPO). At regular intervals during a rollout, for each observation v_t with current substep instruction l_t^{sub}:

1. Compute the policy's action under the correct instruction: $a^+ = \pi_\theta(\cdot \mid v_t, l_t^{\text{sub}})$.
2. Sample a *wrong* substep instruction $\tilde{l}$ from the same task suite but a different task, and compute: $a^- = \pi_\theta(\cdot \mid v_t, \tilde{l})$.
3. Measure the behavioural divergence:

$$r_t^{\text{contrast}} = \min\left(\frac{\|a^+ - a^-\|_2}{\sqrt{d}},\ 1\right), \tag{12}$$

where $d = \dim(\mathcal{A})$.

The total reward for a trajectory is:

$$R = w_1 \cdot R_{\text{task}} + w_2 \cdot \bar{R}_{\text{contrast}}, \tag{13}$$

where $\bar{R}_{\text{contrast}} = \frac{1}{M}\sum_{m=1}^{M} r_{t_m}^{\text{contrast}}$ is the average over M sampled checkpoints.

5.3 Information-Theoretic Interpretation

The contrastive reward directly optimises for high $I(A; L \mid V)$ in the policy's *induced* distribution. Specifically, maximising $\mathbb{E}[\|a^+ - a^-\|]$ under paired instructions is equivalent to maximising the expected divergence:

$$\mathbb{E}_V\left[D\Big(\pi_\theta(A \mid V, L^{\text{sub}}) \parallel \pi_\theta(A \mid V, \tilde{L})\Big)\right], \tag{14}$$

where D is a divergence measure induced by the L_2 norm on the action space. This quantity is a lower bound on $I(A; L \mid V)$ under the policy, ensuring that the RL stage *actively increases* the mutual information between the instruction and the action, conditioned on vision.

5.4 Combined Effect: APD + Contrastive RL

The two components address complementary failure modes:

Component	Mechanism	Information-Theoretic Effect
APD (SFT)	Increases $H(L^{\text{sub}} \mid V)$ in the training distribution	Raises the *upper bound* on $I(A; L \mid V)$, making instruction grounding *possible*
Contrastive RL	Maximises behavioural divergence under different instructions	Pushes the *actual* $I(A; L \mid V)$ toward the raised upper bound, making instruction grounding *realised*

Without APD, the upper bound $H(L \mid V) \approx 0$ constrains the system regardless of the RL signal. Without contrastive RL, the SFT objective permits but does not enforce instruction sensitivity. The combination ensures both a necessary condition (sufficient instruction entropy) and an explicit optimisation pressure (contrastive divergence).

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6. Summary of Theoretical Results

Theorem (Informal). Let π_θ be a VLA policy trained on expert demonstrations. If the language instruction L has low conditional entropy given vision V—i.e., $H(L \mid V) \approx 0$—then:

1. The information gain from language is negligible: $I(A; L \mid V) \leq H(L \mid V) \approx 0$.
2. The optimal policy is approximately invariant to L: $\pi^*(A \mid V, L) \approx \pi^*(A \mid V)$.
3. Gradient-based training drives the language pathway's parameters toward a dormant state: $\|\nabla_{\theta_L} \mathcal{L}\| \approx 0$.
4. Replacing fixed instructions with time-varying substep instructions raises $H(L^{\text{sub}} \mid V) \gg 0$, restoring non-trivial gradients.
5. Contrastive RL post-training further maximises the realised $I(A; L^{\text{sub}} \mid V)$ in the policy's induced distribution.